MIL-T-50606A(MU)
21 July 1975
SUPERSEDING
MIL-T-50606(MU)
30 November 1971

MILITARY SPECIFICATION

TESTER-RECEIVER TRANSMITTER, LASER: 10559600

WARNING - EYE HAZARD

EXTREME CAUTION MUST BE OBSERVED IN OPERATING AND TESTING THE RECEIVER-TRANSMITTER. IF THE LASER LIGHT BEAM ENTERS THE HUMAN EYE, THE RETINA OF THE EYE CAN BE BURNED. THE HIGH ENERGY DENSITY OF THE BEAM CAN ALSO PRODUCE SUCH DAMAGE AT EXTREME RANGE AND BY REFLECTION FROM ANY SHINY SURFACE SUCH AS GLASS, METAL, PAINT, ETC. IN VIEW OF THE POSSIBILITY OF PRODUCING SERIOUS EYE DAMAGE, THE RECEIVER-TRANSMITTER SHOULD NOT BE OPERATED UNLESS THE TARGET AREA HAS BEEN EVACUATED.

1. SCOPE

1.1 This specification covers one type of electro-optical unit known as Tester-Receiver Transmitter, Laser: 10559600 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment: Acquisition, Maintenance, and Disposition of
MIL-R-14865	Receiver-Transmitter, Laser: 10559281
MIL-C-45662	Calibration System Requirements

FSC-4931

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10559600	Tester-Receiver Transmitter, Laser

PACKAGING DATA SHEET

10559600	Packaging of Tester-Receiver Transmitter, Laser: 10559600

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The tester shall be manufactured in accordance with Drawing F10559600 and drawings pertaining thereto.

3.2 General specification.- The tester shall meet the following requirements, where applicable, of MIL-E-13926:

(a) Order of precedence.

(b) Dimensions and tolerance.

(c) Part identification and marking.

3.2.1 Electronic workmanship.- The tester shall meet the workmanship requirements of Drawing F10559600 and drawings pertaining thereto.

3.3 Performance.- Unless otherwise specified, the tester shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (°F) and 90°F.

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3.3.1 Power and signals.- The tester shall perform as specified herein when the auxiliary equipment is connected, the inputs of Table I are applied, and the connections of Figure 1 are made (AN/VVS-1 only).

TABLE I

Item	Condition	Characteristic	Connections
1.	Power sources		Connected between:
1.1	Tank power	24 ±2 Vdc	6A7J1-N(+) and 6A7J1-N(-)
1.2	+29V	29 ±2.5 Vdc	6A7J1-H(+) and 6A7J1-N(-)
1.3	+15V	15 ±1.2 Vdc	6A7J1-J(+) and 6A7J1-N(-)
1.4	+5V	5 ±0.4 Vdc	6A7J1-K(+) and 6A7J1-N(-)
1.5	-5V	-6 ±0. 75 Vdc	6A7J1-L(-) and 6A7J1-N(+)
3.	Auxiliary equipment		
3.1	Laser Range Finder AN/VVS-1 or equivalent		Secured to receiver transmitter tester
	a) Laser beam	50 ±5 mJ	
	b) Extinction	38 ±1 dB	

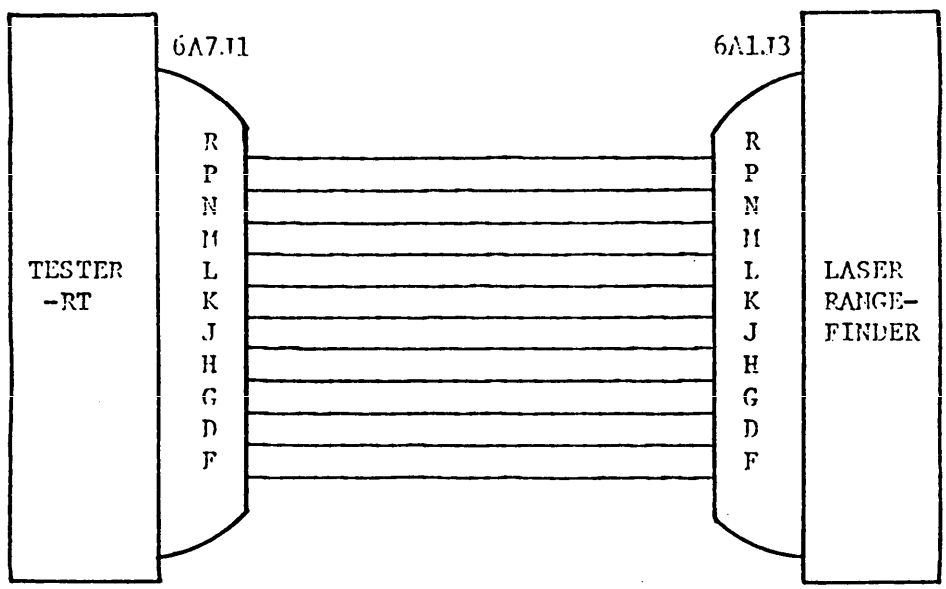


FIGURE 1. Tester to LRF Connections

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3.3.2 Modes of operation.

3.3.2.1 MODE SELECT Switch.

3.3.2.1.1 XMTR position.- When the mode select switch is turned to the furthest counterclockwise position the XMTR mode indicating lamp shall be lit.

3.3.2.1.2 OFF position .- When the mode select switch is turned to the mid-position the OFF mode indicating lamp shall be lit.

3.3.2.1.3 RCVR position.- When the mode select switch is turned to the furthest clockwise position the RCVR mode indicating lamp shall be lit.

3.3.2.2 Transmitter test mode.- When the mode select switch is in the position which lights the XMTR lamp, the following specifications shall be met.

3.3.2.2.1 RESET-FIRE switch.

3.3.2.2.1.1 RESET position.- With the RESET-FIRE switch pushed down to the RESET position, the specified d-c voltage shall appear at the terminal listed in Table II, referenced to 6A7J1-N (ground).

TABLE II

Output Pin	Output Voltage
6A7J1-P	0.2 ±0.2 Vdc

3.3.2.2.1.2 FIRE position.- When the RESET-FIRE switch is pulled out and up to the FIRE position, the specified d-c voltages shall appear at the terminals listed in Table III, referenced to ground.

TABLE III

Output Pins	Output Voltages
6A7J1-R	1.3 ±0.7 Vdc
6A7J1-P	4 ±1.5 Vdc

3.3.2.2.1.3 RCVR-XMTR STATUS meter reading.- Meeting condition 3.3.2.2.1.2 shall cause a deflection of the indicating needle of the RCVR-XMTR STATUS meter within the red scale.

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3.3.2.3 Receiver test mode.- When the mode select switch is in the RCVR position, the RCVR lamp shall cause a deflection of the indicating needle of the RCVR-SITP STATUS meter within the white scale. A momentary ground applied to pin 6A7J1-D shall not cause any portion of the indicating needle of the RCVR-XMTR STATUS meter to deflect into the extreme left white border of the meter scale.

3.3.3 Environmental.

3.3.3.1 Storage temperature.- The tester shall meet the requirements of 3.3.2.1 through 3.3.2.3 at ambient temperature (60°F to 90°F) after exposure and thermal stabilization at -65°F and +160°F.

3.3.3.2 Operating temperature.- The tester shall meet the requirements of 3.3.2.1 through 3.3.2.3 while exposed to and thermally stabilized at 40°F and 100°F.

3.3.3.3 Vibration.- The tester shall meet the requirements of 3.3.2.1 through 3.3.2.3 after being exposed to the following vibratory conditions:

a.	Amplitude:	Simple harmonic motion at 0.03 inch amplitude (0.06 inch total excursion) maximum.
b.	Acceleration:	1.5 g maximum.
c.	Sweep Time:	10 Hz to 55 Hz to 10 Hz in 5 minutes, two sweeps in each of three mutually perpendicular axes.

3.3.3.4 Shock.- The tester shall meet the requirements of 3.3.2.1 through 3.3.2.3 after being exposed to a total of 6 shock impulses, one in each of three mutually perpendicular axes. Each shock impulse shall be a sawtooth or half sine-wave with a time duration of 11 milliseconds. The peak amplitude of each shock impulse shall be 30 g.

3.3.3.5 Immersion.- The tester shall show no evidence of leakage (gas bubbles from inside the unit) when immersed in water, at 110 ±10°F, to a depth of 2 feet maximum for a 30 minute observation period. Prior to immersion the tester shall have been pressurized to an internal pressure of 10 ±1 psig.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three test sets. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables IV, V and VI herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables IV, V, and VI shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests.- The requirements and tests of Table IV shall be inspected on 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60°F to 90°F).

TABLE IV

(100 Percent Inspection)

Characteristic	Requirement	Test Procedure
101 Mode Select switch on XMTR Position	3.3.2.1.1	4.6.2.1.1
102 MODE SELECT Switch on OFF Position	3.3.2.1.2	4.6.2.1.2
103 MODE SELECT Switch on RCVR Position	3.3.2.1.3	4.6.2.1.3
104 RESET-FIRE Switch in RESET Position	3.3.2.2.1.1	4.6.2.2.1
105 RESET-FIRE Switch in FIRE Position	3.3.2.2.1.2	4.6.2.2.2
106 Transmitter Status	3.3.2.2.1.3	4.6.2.2.3
107 Receiver Status	3.3.2.3	4.6.2.3.1

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4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One tester shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and test in Table V.

TABLE V

Control Test	Requirement	Test Procedure
301 Fabrication	3.1	Applicable drawings-Visual
302 General specification	3.2	MIL-F-13926 - Visual

4.4.2 Environmental.- Three testers shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table VI.

TABLE VI

Control Test	Requirement	Test Procedure
303 Storage temperature	3.3.3.1	4.6.3.2
304 Operating temperature	3.3.3.2	4.6.3.2
305 Vibration	3.3.3.3	4.6.3.3
306 Shock	3.3.3.4	4.6.3.4
307 Immersion	3.3.3.5	4.6.3.5

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted item has been approved (see 4.3.3).

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4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SOAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" provisions of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2, the standard test equipment listed in Table VII shall be utilized in performing the applicable test of 4.6.

4.6 Test methods and procedures.

4.6.1 Test conditions.- Unless otherwise specified, the test conditions shall be in accordance with the "Test Facilities" provisions of MIL-F-13926.

4.6.1.1 Initial conditions.- Connect the auxiliary equipment and power to the receiver-transmitter as specified in Table I. Make the electrical connections shown in Figure 1. Perform the calibration adjustment of 4.6.1.2.

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TABLE VII

Nomenclature	Description
1. D-C Power Supply	Plus 29 ±1.5 volts Ripple: 0.1 volt rms maximum
2. D-C Power Supply	Plus 24 ±2 volts Ripple: 0.1 volt rms maximum
3. D-C Power Supply	Plus 15 ±0.7 volts Ripple: 0.1 volt rms maximum
4. D-C Power Supply	Plus 5 ±0.2 volt Ripple: 0.1 volt rms maximum
5. D-C Power Supply	Minus 6 ±0.2 volt Ripple: 0.1 volt rms maximum
6. D-C Voltmeter	Range: 0 to 50 volts Accuracy: ±0.3%
7. Oscilloscope with Camera	Frequency Response: dc to 150 MHz Sweep Time: 0.01 μs/cm Sensitivity: 5 mV/cm Camera Lens: f/1.3 (Tektronix C40 or equivalent)
8. Receiver-Transmitter, LASER or equivalent*	MIL-R-14865 (MI)

NOTE: This item shall be calibrated in accordance with MIL-C-45662, and approved by Contracting Officer, Frankford Arsenal, Phila., PA prior to use as test equipment.

4.6.1.2 Tester calibration adjustment.- Turn the MODE SELECT switch counterclockwise to the SMTR position. Remove sealing screw from R3 access port. Set the RESET-FIRE switch momentarily to the down (RESET) position, then momentarily to the up (FIRE) position. Adjust potentiometer R3 until the indicating needle of the RCVR-SMTR STATUS meter is deflected to the midpoint of the red scale, disregarding meter overshoot. Turn the MODE SELECT switch counterclockwise (RCVR) position. Remove sealing screw from R9 access port. Adjust R9 until the indicating needle of the RCVR-SMTR STATUS meter is deflected to the midpoint of the white scale. Turn the MODE SELECT switch counterclockwise to the OFF position. Replace sealing screws in the R3 and R9 access ports.

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4.6.2 Modes of operation.

4.6.2.1 MODE SELECT switch.

4.6.2.1.1 XMTR position test.- Turn the MODE SELECT switch to its furthest counterclockwise position. Visually check the XMTR lamp. The requirements of 3.3.2.1.1 shall be met.

4.6.2.1.2 OFF position test.- Turn the MODE SELECT switch to its midposition. Visually check the OFF lamp. The requirements of 3.3.2.1.2 shall be met.

4.6.2.1.3 RCVR position test.- Turn the MODE SELECT switch to its furthest clockwise position. Visually check the RCVR lamp. The requirements of 3.3.2.1.3 shall be met.

4.6.2.2 Transmitter tests.- Turn the MODE SELECT switch to the XMTR position.

4.6.2.2.1 RESET-FIRE switch in RESET position test.- Push the RESET-FIRE switch down to the RESET position. Measure the voltages listed in Table II. The requirement 3.3.2.2.1.1 shall be met.

4.6.2.2.2 RESET-FIRE switch in FIRE position test.- Pull the RESET-FIRE switch out and up to the FIRE position. Measure the voltages listed in Table III. The requirements of 3.3.2.2.1.2 shall be met.

4.6.2.2.3 Transmitter status test.- Set the RESET-FIRE switch to the FIRE position. Visually check for a deflection of the indicating needle of the RCVR-XMTR STATUS meter. The requirements of 3.3.2.2.1.3 shall be met.

4.6.2.3 Receiver tests.- Turn the MODE SELECT switch to the RCVR position.

4.6.2.3.1 Receiver status test.- Visually check for a deflection of the indicating needle of the RCVR-XMTR STATUS meter. The requirements of 3.3.2.3 shall be met.

4.6.3 Environmental tests.

4.6.3.1 Tester purging and pressurization.- With the unit attached to a nitrogen cylinder, complete the following purge operations:

a. With the nitrogen exhaust port screw removed, pressurize the tester at an inlet pressure of 14.5 ±1 psig and allow it to purge for a minimum period of 1 minute, then depressurize to ambient pressure. Repeat this process at least 4 times.

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b. Repressurize the tester with an inlet pressure of 14.5 ±1 psig and allow it to purge for a minimum of 20 minutes; then reduce the inlet pressure to less than 3 psig and reinstall the nitrogen exhaust screw. Repressurize again to a pressure of 10 ±1 psig.

4.6.3.2 Temperature.- The test chamber shall be in accordance with the applicable "Test Facilities" provisions of MIL-F-13926. The test equipment shall be positioned outside the test chamber and be capable of presenting the necessary signals to the tester in the temperature chamber. Perform this test in accordance with Procedure I of MIL-F-13926 except that the low storage temperature shall be -65°F in lieu of -80°F and the operating conditions shall be +40°F and +100°F in lieu of those specified in Table IV therein. The rate of temperature change during cycling shall not be greater than 7°F per minute. At each of the specified operating temperatures (including ambient), perform the tests of 4.6.2.1.1 through 4.6.2.3.1 to determine compliance with 3.3.3.1 and 3.3.3.2.

4.6.3.3 Vibration.- Perform this test utilizing the apparatus specified in Method 514.1 of MIL-STD-810 and in accordance with the vibratory conditions specified in 3.3.3.3. Following vibration exposure, examine the tester for structural failures, loose hardware, wire breakage, and abrasion and perform the tests of 4.6.2.1.1 through 4.6.2.3.1 to determine compliance with 3.3.3.3.

4.6.3.4 Shock.- Perform this test utilizing apparatus specified in Method 516.1 of MIL-STD-810, in accordance with Procedure I therein and with the conditions specified in 3.3.3.4. Following exposure to shock, examine the tester for structural failures, loose hardware, and wire breakage and perform the tests of 4.6.2.1.1 through 4.6.2.3.1 to determine compliance with 3.3.3.4.

4.6.3.5 Immersion.- Connect the nitrogen charging valve on the tester to a cylinder of nitrogen and pressurize the tester to 10 ±1 psig. Observe the tester for evidence of leakage (gas bubbles) for a period of 30 minutes to determine compliance with 3.3.3.5.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10559600. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The tester is a unit of the laser range finder special tools and test equipment.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable levels of preservation, packaging, and packing.

d. Applicable stock number.

e. Requirement for First Article Submission.

Custodian:

Army - MU

Preparing Activity:

Army - MU
Project No. 4931-A129

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